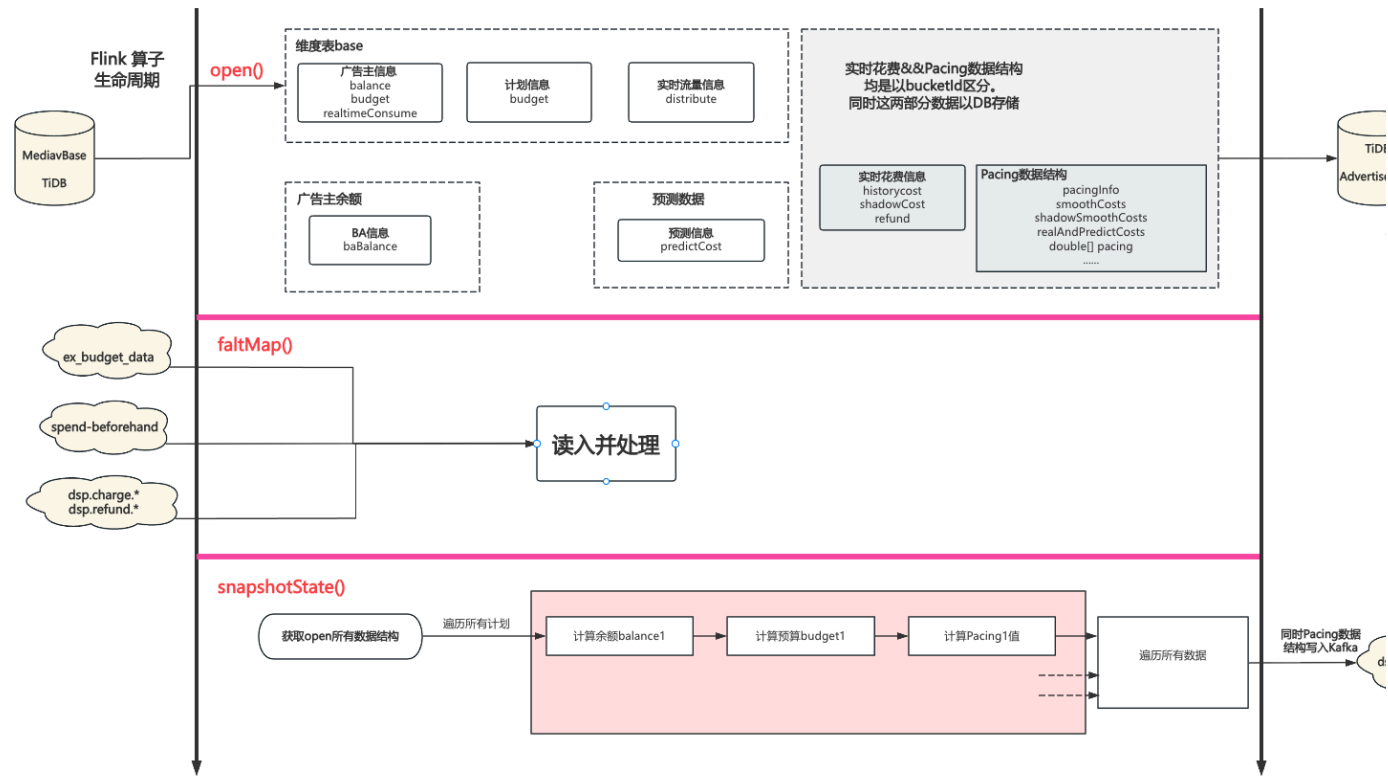


Pacing项目全逻辑梳理

由 朱明创建, 最后修改于三月 04, 2026 144 views since 10 Oct 2024

- 一、项目背景介绍
 - Pacing项目的结构
- 二、Pacing项目细节分析
 - 1、流量预测分布
 - 2、实现精确的调节因子
 - 广告主&&计划属性调整
 - 广告主如何主动加快投放
 - equalbudgetflag设置为1
 - equalbudgetflag设置为0
 - 3、快速结束
 - 4、慢开始



一、项目背景介绍

预算控制在广告系统中的作用就是该如何合理花掉广告主的预算。

1、投放过慢

显而易见，广告主无法投放广告，影响收益

2、投放过快

在实际中经常会出现广告主预算消耗过快的问题，这会导致广告主早早退出竞价，不仅会影响广告主体验，也会导致整个广告生态的竞争力下降

Linked行业主流方案

<https://github.com/wzhe06/Ad-papers/blob/master/Budget%20Control/Budget%20Pacing%20for%20Targeted%20Online%20Advertisements%20at%20LinkedIn.pdf>

算法的主要思想就是令每个 **campaign**（推广计划）的消耗趋势与其曝光变化趋势基本保持一致，以天为时间单位，**campaign** 为预算控制单位，首先为每个 **campaign** 预测出其在当天的曝光情况；

然后基于其曝光情况，在当前时间片，假如 已消耗/当天预算 的比例大于 已曝光/预测的总曝光 的比例，则说明预算已经消耗过快，需要减小消耗的速度，反之则要加快消耗的速度

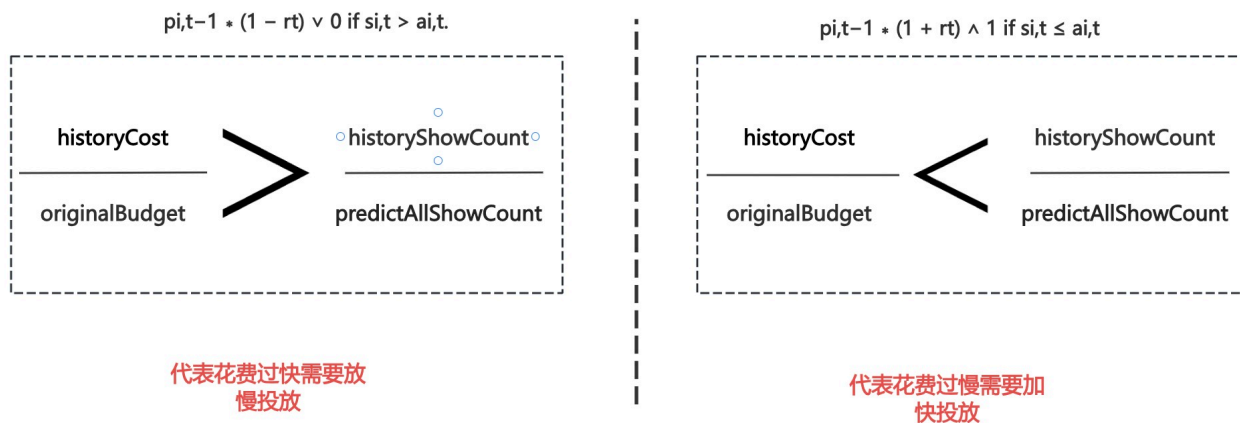
核心公式如下：

where $f_{i,T}$ is the forecasted total volume of eligible impressions for campaign i for the entire day. We apply a filter during each ad auction. For each eligible auction a campaign is allowed to participate with probability $p_{i,t}$, the pass through rate (PTR). The value of p_{it} is updated at the start of each time window by

$$p_{i,t} = \begin{cases} p_{i,t-1} * (1 + r_t) \wedge 1 & \text{if } s_{i,t} \leq a_{i,t} \\ p_{i,t-1} * (1 - r_t) \vee 0 & \text{if } s_{i,t} > a_{i,t}. \end{cases} \quad (2)$$

Here $0 < r_t < 1$ is an adjustment rate.

实际含义：

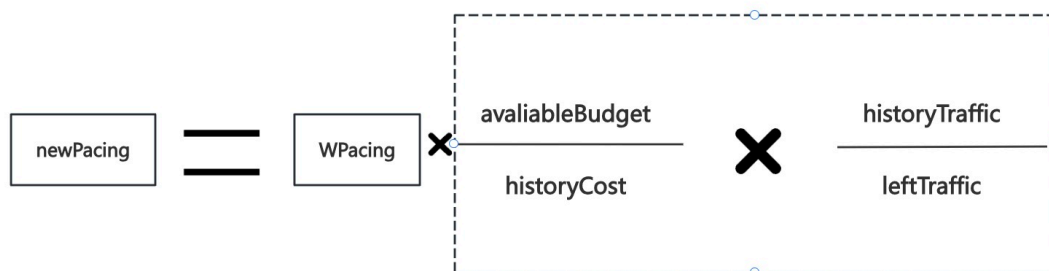


Pacing项目的结构

Paccing项目借鉴并优化了控制细节

Pit 胜出成功率 = 当天截止目前的 加权WPaicng值

```
newWK = Math.max(Constants.PACING_MIN_WK, oldK * ((double) mintraffic / historyTraffic) + oldWK * ((double) (historyTraffic - mintraff
```

$$Rt \text{ 调节因子} = ((\text{double}) \text{ availableBudget} / \text{historyCost}) * ((\text{double}) \text{ historyTraffic} / \text{leftTraffic})$$


下面两个值为实际值

availableBudget 当天剩余可用预算

histortCost 当天已经花费

下面两个值是预测值

histryTraffic 当天已经消耗流量

leftTraffic 当天剩余可用流量

二、Pacing项目细节分析

五大原则 Linked

原则	说明
Update Frequency	分钟
Forecasting	细粒度预测曝光数
Adjustment Rate	Rt调节因子
Slow Start	平滑开始

原则	说明
Fast Finish	快速结束

1、流量预测分布

Linked 的实现方式

2.1.2 Forecasting

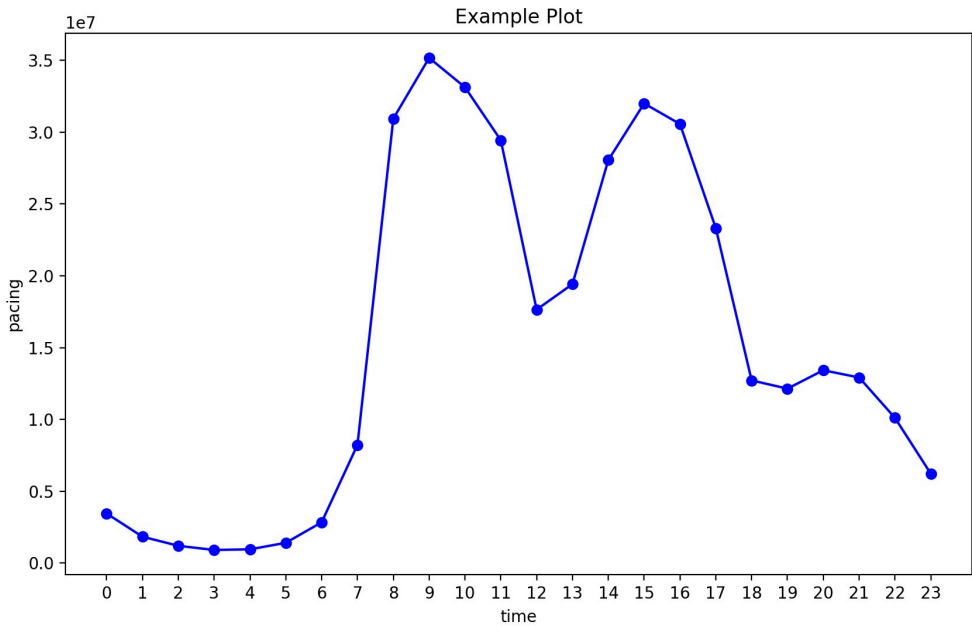
Forecasting the number of eligible impressions for campaigns plays an important role in our algorithm. It is used to determine the allocation curve for each campaign. Inaccurate forecasts may cause unnecessary throttling and may result in budget not being used up for some good campaigns.

We use a tool that we built at LinkedIn using the method described in [3]. Note that, in order to forecast the number of eligible impressions for campaign i , we need to forecast the number of impressions from the member segment $\mathcal{S}_i$, which is the target audience for campaign i . We maintain a large list of important member segments which we call *base profiles*. For each base profile we update the historical time series data of number of ad impression opportunities. We use an ensemble statistical time series model to fit the up-to-date time series data for each of these base profiles daily. We obtain forecasts for each targeted segment $\mathcal{S}_i$ using correlation models described in [3].

Pacing目前预测的粒度是dealMode

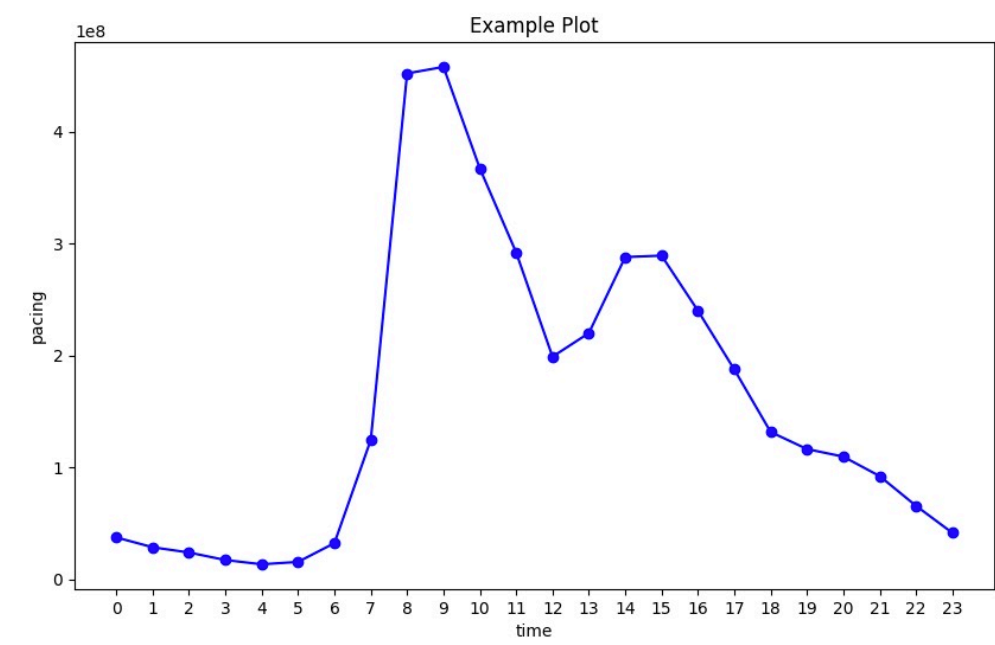
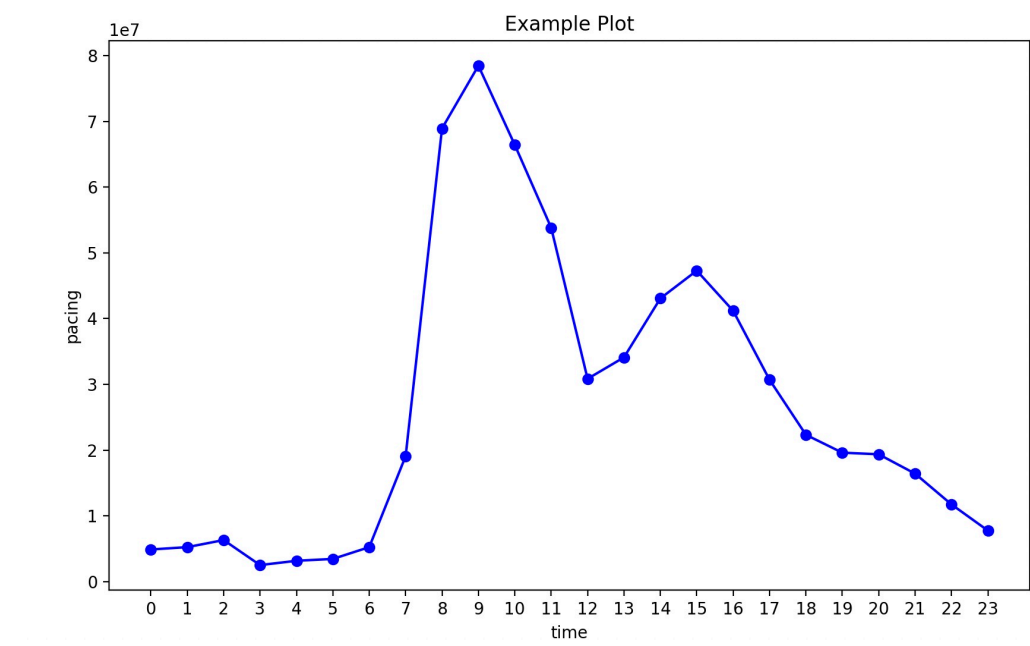
dealmode	sum(traffic)
1	419661937
2	446703445
3	1950500
0	367996533

dealMode=0 的 Traffic流量预测全天小时分布图



以一天的曝光数据作为指标，可以发现和Pacing的traffic数据保持一致

```
1 SELECT dealmode, hour ,sum(show_count)
2 FROM d_realtime_qihu
3 WHERE date='2024-10-11'
4 GROUP BY dealmode, hour
```



2、实现精确的调节因子

Linked实现方式

Rt因子为常量，通过调节选择公式， $T \rightarrow T+1 \rightarrow T+2$ 自适应调整Pit

2.1.3 Adjustment Rate

We set the adjustment rate r_t at a constant 10%. This naive choice of the adjustment rate is not only the easiest to implement, but is also a very robust one. A more theoretically sound algorithm should involve the derivative $\partial s_{i,t}/\partial t$, the rate of budget spend while computing the PTR; a campaign that spends budget very fast needs to be throttled more whereas a campaign spending budget just a little bit faster than the allocation curve needs to be throttled less. We chose the naive form for two main reasons. First, an estimate for $\partial s_{i,t}/\partial t$ is likely to be noisy. The budget spend curve is not smooth, especially for cost-per-click campaigns for whom it changes only when there is a click which is significantly rare compared to the impressions. Second, we adjust the PTR frequently and even if we are not at the ideal PTR at any time point we can reach there fast.

Pacing的实现方式

广告主计划属性调整

广告主属性

广告主维度	说明
budgetflag	预算限制标志(0:有预算限制, 1:无预算限制); 为1可以使用广告主全部余额
is_over_delivery	是否受到恶意超投限制 0: 关闭限制 1: 开启限制
resource_type	用来标示是360投放还是聚效投放,qihu:360中小;qihu_ka:360ka;360奇迹
status	广告主状态(-3:拒绝,-2:试用(审),-1:试用,0:冻结,1:有效,2:余额不足,3: 360广告主预算受限不足)
disabled	广告状态, 0: 有效,1: 无效, 2: 删除

计划属性

计划维度	说明
budgetflag	预算限制标志(0:有预算限制, 1:无预算限制)
budget	每日广告活动预算(单位为分)
equalbudgetflag	预算均匀投放限制标志(0:有设置, 1:无设置)
show_budget_flag	每日展现预算限制标志(0:无限制, 1:有限制)
is_offline_screen	是否融屏广告 0:否, 1:是
displaytype	
status	花费状态(0:余额或预算不足,1:正常投放)
disabled	订单状态, 此状态由用户在列表中修改, 0: 已启用, 1: 暂停, 2: 删除

广告主如何主动加快投放

计划维度表的字段

计划标记	含义	实际分类	数量级别
equalbudgetflag	预算均匀投放限制标志(0:有设置, 1:无设置)	0	10747
		1	4801

equalbudgetflag设置为1

leftTraffic 变为 5分钟Traffic, 提高系数

availableBudget

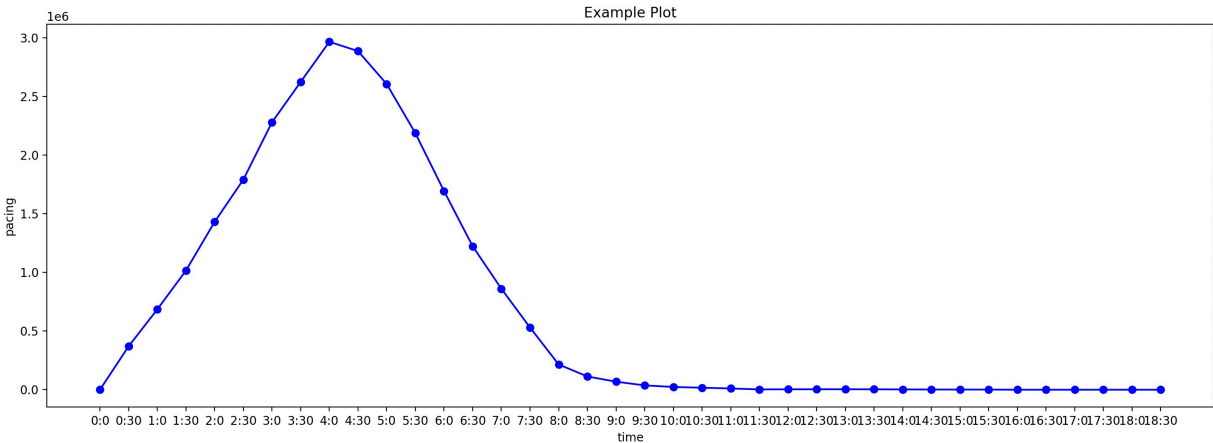
historyCost

×

historyTraffic

5minTraffic

4495750计划的newPacing值



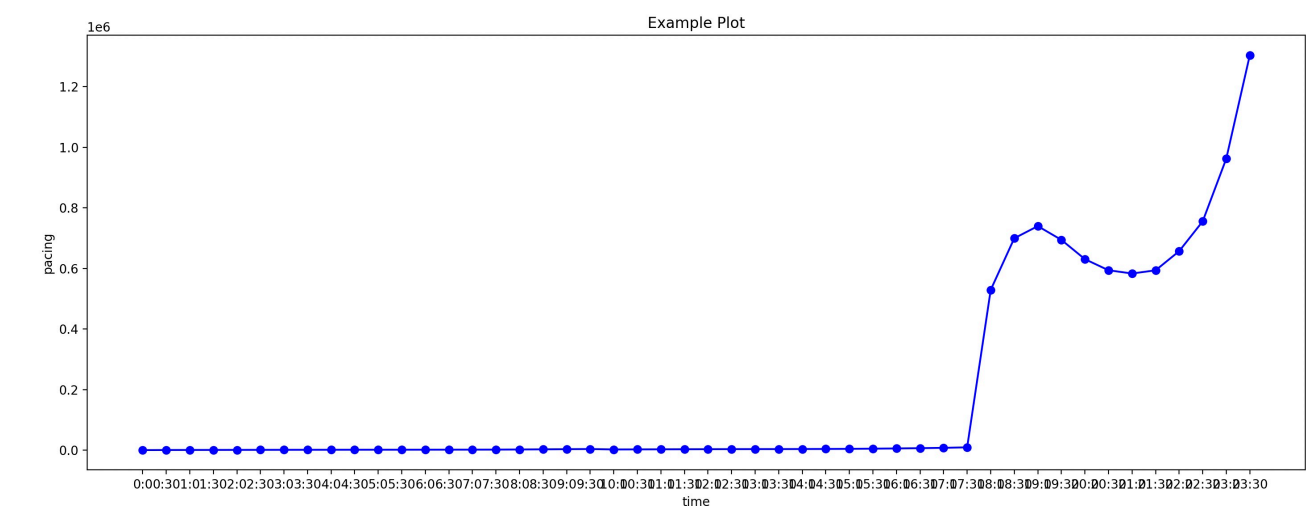
快开始、快结束

time, bucketPacing, bucketWPacing, bucketHistoryCost, bucketAvailableBudget, totalTraffic, leftTraffic, pidValue

('2024-10-09 00:00:12', 100, 100, 713, 0, 0, 0, 0.0)
('2024-10-09 00:01:12', 70, 70, 5523, 359993928, 0, 0, 0.0)
('2024-10-09 00:02:12', 140, 70, 104583, 359894376, 36799653, 36777354, 142715.5987679248)
('2024-10-09 00:03:12', 210, 87, 666087, 359333886, 36799653, 36770054, 38175.20802313588)
('2024-10-09 00:04:12', 280, 111, 1144548, 358855452, 36799653, 36762934, 35915.08941971016)
('2024-10-09 00:05:12', 350, 139, 1395187, 358604448, 36799653, 36755580, 42889.450436301646)
('2024-10-09 00:06:12', 420, 168, 1561569, 358438263, 36799653, 36748410, 55357.08663702396)
('2024-10-09 00:07:12', 490, 200, 1705937, 358293895, 36799653, 36741087, 67227.98892162474)
('2024-10-09 00:08:12', 560, 232, 1866842, 358132990, 36799653, 36733829, 80760.82566179306)
('2024-10-09 00:09:12', 630, 263, 2005668, 357991719, 36799653, 36726750, 96911.83760022215)
('2024-10-09 00:10:12', 700, 295, 2133729, 357866046, 36799653, 36719838, 114511.00306231581)
('2024-10-09 00:11:12', 770, 328, 2283270, 357715954, 36799653, 36712770, 126490.70729449908)
('2024-10-09 00:12:12', 840, 361, 2437639, 357562136, 36799653, 36705821, 143091.61096686262)
('2024-10-09 00:13:12', 910, 394, 2568254, 357431671, 36799653, 36698883, 159368.89903638774)
('2024-10-09 00:14:12', 980, 426, 2761031, 357238775, 36799653, 36692151, 176230.5273834626)
('2024-10-09 00:15:12', 1024, 458, 2990439, 357009267, 36799653, 36685467, 187160.79156745708)

结束时间	pacing	wPacing	historyCost	availableBudget	wPacing*PID
2024-10-09 15:52:12	1024	1021	355278628	4072398	1081.611210675429
2024-10-09 15:53:12	897	1021	355321945	3397175	897.0545395514062
2024-10-09 15:54:12	697	1021	355371072	2561531	697.8869264742532
2024-10-09 15:55:12	10	1020	355411434	1914352	10.887414504374163
2024-10-09 15:56:12	7	1018	355464032	1368864	7.816560897602807
2024-10-09 15:57:12	4	1016	355512425	775132	4.444249748341788

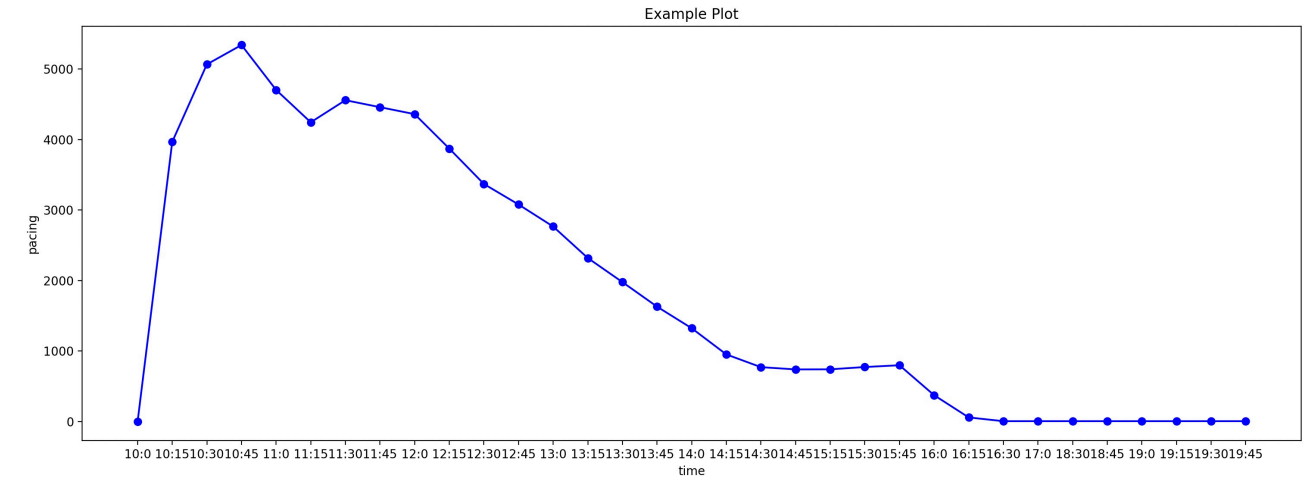
equalbudgetflag设置为0
计划4487989的newPaicng值



慢开始

时间	pacing	wPacing	historyCost	availableBudget			wPacing*PID
2024-10-09 03:16:12	1017	1021	8854420	990557306	36799653	36121963	1017.5619515600051
2024-10-09 03:17:12	1021	1021	8864256	990588134	36799653	36120404	1021.4007090114637
2024-10-09 03:18:12	1024	1021	8877040	990621188	36799653	36118842	1024.924384863046

计划4488641的newPaicng值



慢结束

('2024-10-09 14:12:12', 1024, 976, 67545598, 119670253, 28103499, 17613234, 1029.9214893258604
('2024-10-09 14:13:12', 1005, 976, 68260832, 117255864, 28103499, 17569606, 1005.4161425972162
('2024-10-09 14:14:12', 981, 976, 68959080, 114901118, 28103499, 17525735, 981.8841271542269)
('2024-10-09 14:15:12', 953, 976, 70017177, 112513235, 28103499, 17481950, 953.2722247366805)
('2024-10-09 14:16:12', 904, 976, 71923810, 108953748, 28103499, 17437731, 904.5841441522747)
('2024-10-09 14:17:12', 852, 975, 73908736, 104826889, 28103499, 17393272, 852.3916966379487)
('2024-10-09 14:18:12', 809, 975, 75661975, 101325205, 28103499, 17348643, 809.8331021939509)
('2024-10-09 14:19:12', 806, 974, 75818008, 100470142, 28103499, 17303775, 806.2070097079549)

3、快速结束

Linked实现方式

2.1.5 *Fast Finish*

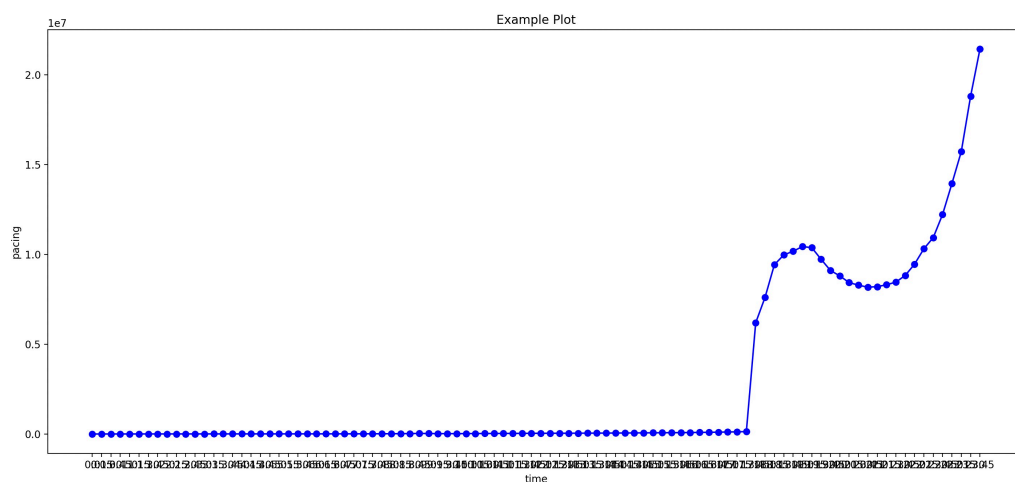
We use statistical models to forecast future supply of impressions for each campaign. The models typically have high accuracy but there are always statistical errors. As a result there may be situations where the algorithm continuously sets lower values of PTR for a campaign than what should have been set. We try to mitigate this issue by *fast finish*- we modify the allocation curve slightly so that the algorithm assumes there will not be any traffic during the last two hours of the day. This way the pacing algorithm attempts to exhaust budget within 22 hours. This in turn gives campaigns that have been wrongly throttled earlier in the day, more opportunities to use up their budget later.

Pacing处理方式

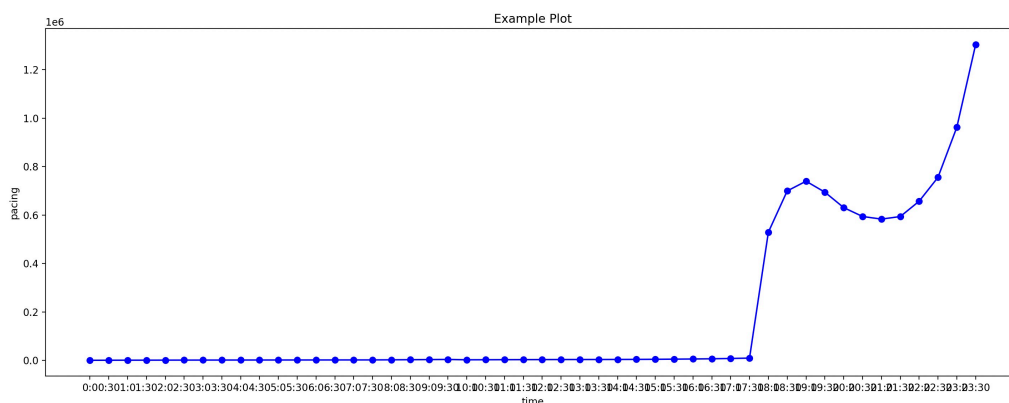
在排期的最后一小时，或者流量的最后20%，开启加速

$$\frac{\text{availableBudget}}{\text{historyCost}} \times \frac{\text{historyTraffic}}{\text{5minTraffic}}$$

计划4486399, ocpc_flag=6,equalbudgetflag=0



计划 4487989, ocpc_flag=6,equalbudgetflag=0



临界点如何处理?

stop 1 minute ahead

在当前可用余额 < 当前预测一分钟花费，则停止投放

4、慢开始

Linked实现方式，类似 $\text{Paicng } 1024 \times 0.1 = 102$ 作为初始值

2.1.4 Slow Start

We always start with a PTR of 10% for all campaigns. We call this a *slow start*. This may be a low starting value for some campaigns and may be a high value for some other campaigns. The advantage of a low starting value is that it gives the system the time to learn about individual campaigns. On one hand, if a campaign is spending its budget too slow then it can open up fully with in 25 minutes. On the other hand if a campaign is spending too fast then having a 10% PTR to start with gives the system more time to reach the correct PTR. For such campaigns starting from a high PTR may result in using up the budget before the system learns that it needs to set a low PTR.

It is sensible to customize the starting value of the PTR for individual campaigns. We aim to do that in future developments where we plan to make the initial value of the PTR for a campaign, a function of the demand and the forecasted supply for the campaign. The demand for a campaign is the number of impressions required to spend the budget of the campaign and the supply is the total number of impressions available for the campaign. It is natural to set a low initial value of PTR for a campaign with a high demand-supply ratio.

Pacing实现方式

```
1 usage  zhuming3
public static List<ColdStartData> constructColdStartExptForBucket() {
    List<ColdStartData> res = new LinkedList<>();
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * Constants.PACING_STOP_BUDGET_BUCKET, pacingInit: 0, pacingAcceleration: 0, pacingMaxCold: 0,
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 30 / 10, pacingInit: 1, pacingAcceleration: 1, pacingMaxCold: 3, pacingMaxNormal: 20));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 50 / 10, pacingInit: 1, pacingAcceleration: 1, pacingMaxCold: 5, pacingMaxNormal: 50));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 100 / 10, pacingInit: 2, pacingAcceleration: 2, pacingMaxCold: 15, pacingMaxNormal: 100));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 300 / 10, pacingInit: 5, pacingAcceleration: 5, pacingMaxCold: 20, pacingMaxNormal: 200));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 500 / 10, pacingInit: 10, pacingAcceleration: 10, pacingMaxCold: 60, pacingMaxNormal: 200));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 1000 / 10, pacingInit: 20, pacingAcceleration: 20, pacingMaxCold: 120, pacingMaxNormal: 300));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 3000 / 10, pacingInit: 30, pacingAcceleration: 30, pacingMaxCold: 150, pacingMaxNormal: 300));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 5000 / 10, pacingInit: 40, pacingAcceleration: 40, pacingMaxCold: 200, pacingMaxNormal: 500));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 10000 / 10, pacingInit: 50, pacingAcceleration: 50, pacingMaxCold: 300, pacingMaxNormal: 800));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 30000 / 10, pacingInit: 60, pacingAcceleration: 60, pacingMaxCold: 400, pacingMaxNormal: 800));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 50000 / 10, pacingInit: 70, pacingAcceleration: 70, pacingMaxCold: 500, pacingMaxNormal: 800));
    res.add(new ColdStartData( budget: Constants.UNIT_ADVERTISER_MONEY_TO_COST * 100000 / 10, pacingInit: 80, pacingAcceleration: 80, pacingMaxCold: 600, pacingMaxNormal: 800));
    return res;
}
```

无标签